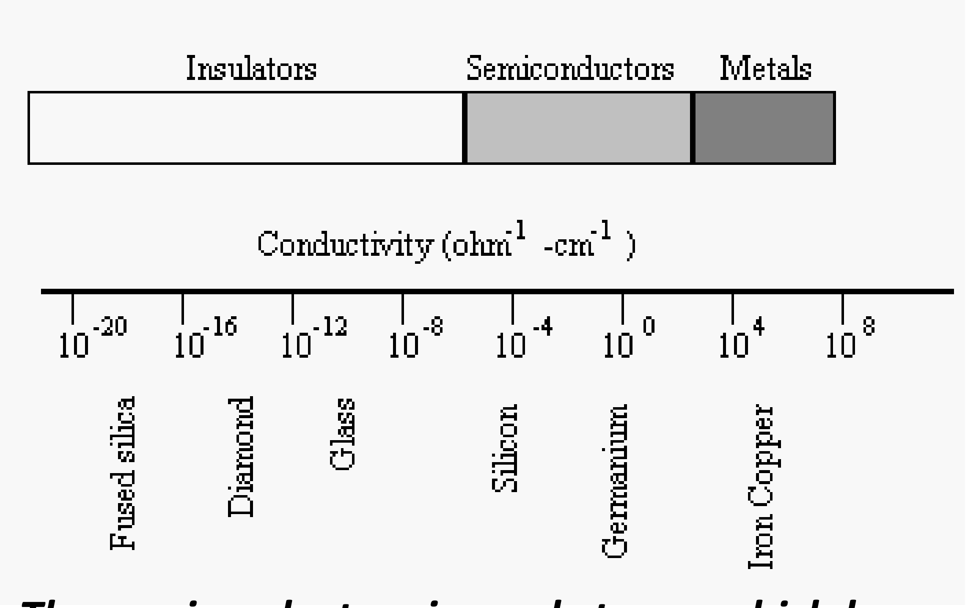
Semiconductor Devices and Applications

Types of Material



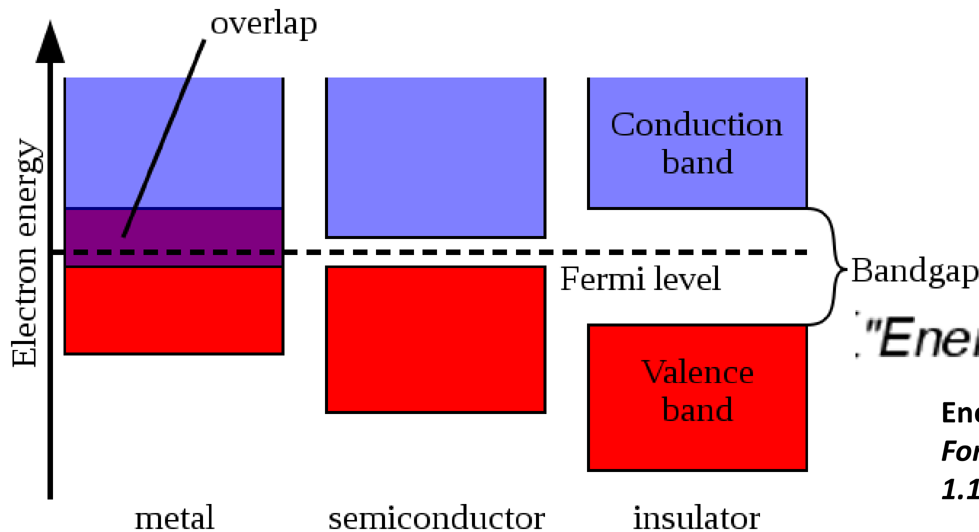
The semiconductors is a substance which has resistivity in-between conductors and insulators.

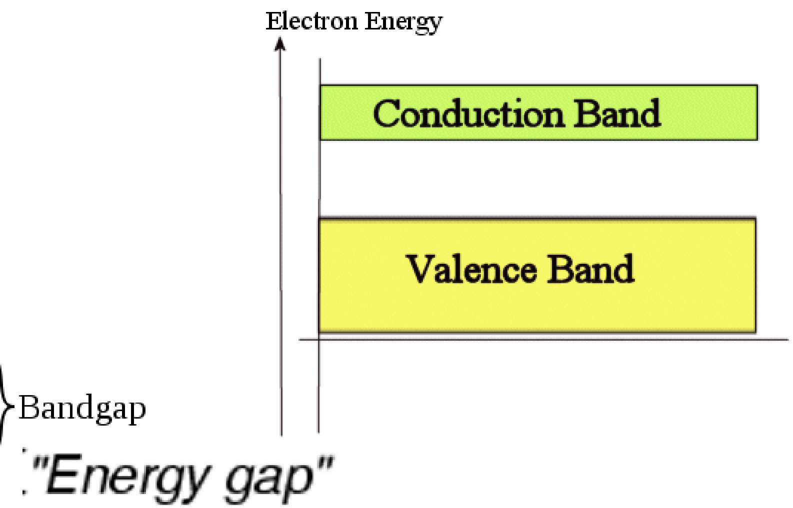
Example: Silicon, Germanium, Selenium, Carbon.

$$R = \rho\frac{L}{A} \quad \text{and} \quad \sigma = \frac{1}{\rho}$$

$$\therefore R = \frac{L}{\sigma A}\ \Omega$$

Types of Material





Energy Gap in Semiconductors

Forbidden energy gap: 0.7 eV for germanium and 1.1 eV for Silicon.

Valence Band: The highest occupied energy band is called the valence band. Most electrons remain bound to the atoms in this band.

Conduction Band: **The area beyond the outermost shell is called the conduction band.** ***The conduction band is the band of orbitals that are high in energy and are generally empty.***

It is the band that accepts the electrons from the valence band.

Properties of Semiconductor:

1: The resistivity of a semiconductor is less than an insulator but higher than a conductor.

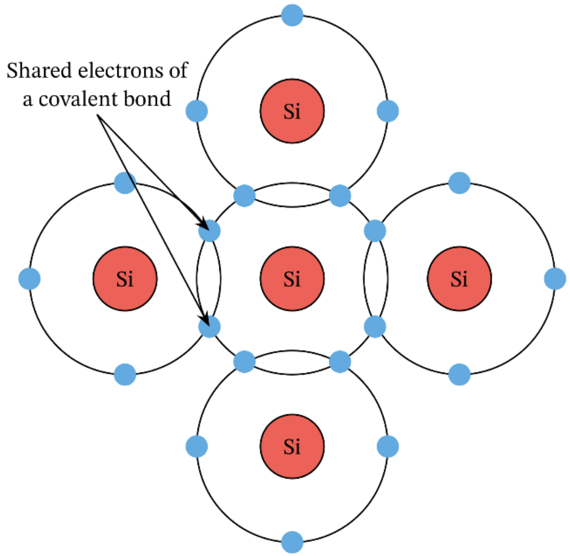
2: Semiconductors show a negative temperature coefficient of resistance. In simple words, the resistance of the semiconductors decreases as the temperature increases and vice-versa.

3: At zero kelvin, semiconductors behave as insulators. As the temperature is increased it works as a conductor

4: The conductivity of the semiconductors increases when impurities are added. The process of adding impurities to semiconductors is called doping.

Bonds in Semiconductors

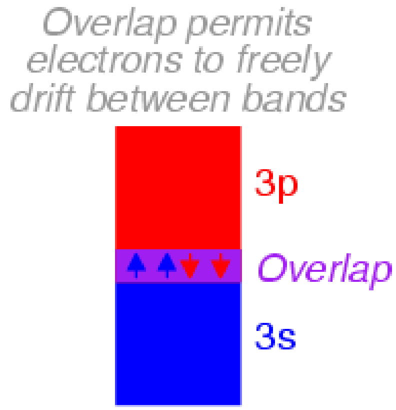
The semiconductors such as silicon and germanium contains covalent bonding. Two adjacent atoms share electron and form a covalent bond.



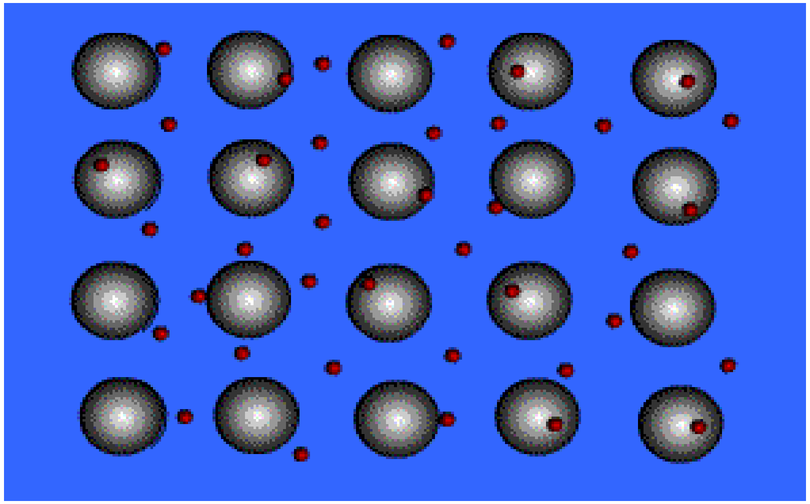
The electrons surrounding each atom in a semiconductor are part of a covalent bond. A covalent bond consists of two atoms "sharing" a pair of electrons.

Conductors

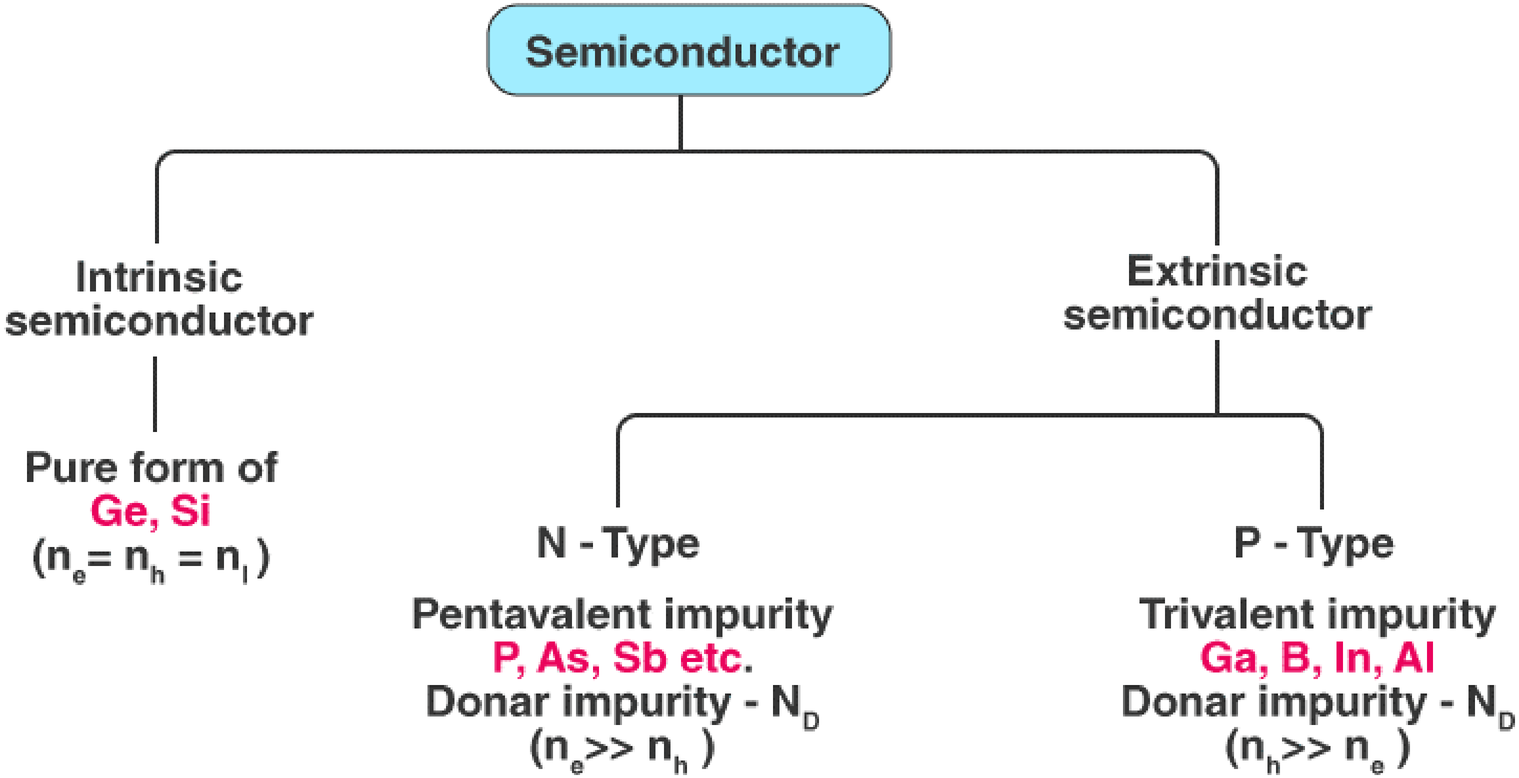
In a conductor, electrons can move freely among these orbitals within an energy band as long as the orbitals are not completely occupied.



Multitudes of atoms in close proximity



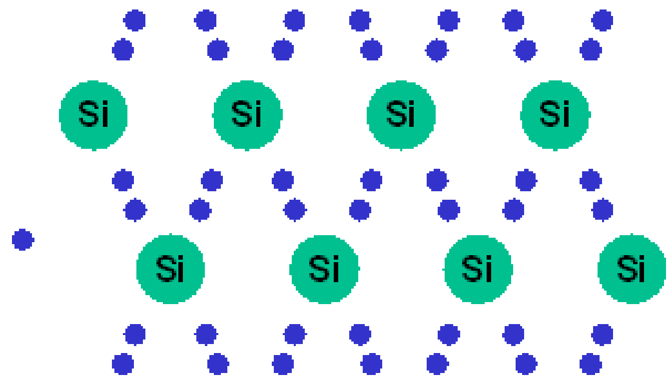
Semiconductor



Semiconductor

Intrinsic Semiconductor:

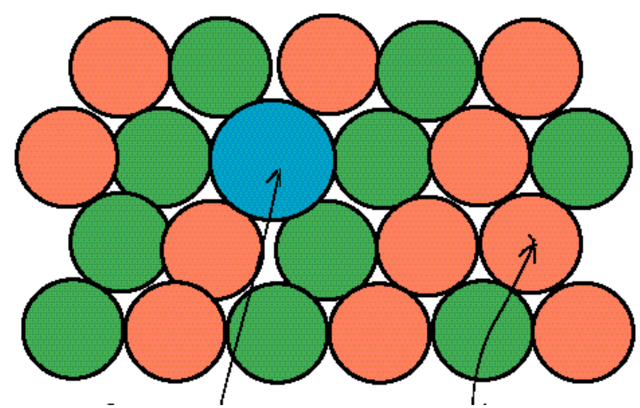
A semiconductor in an extremely pure form is known as an intrinsic semiconductor.

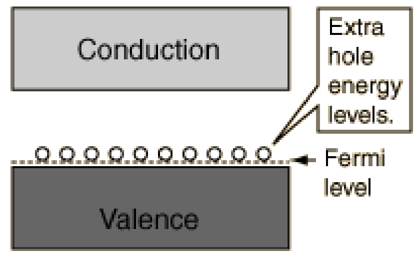


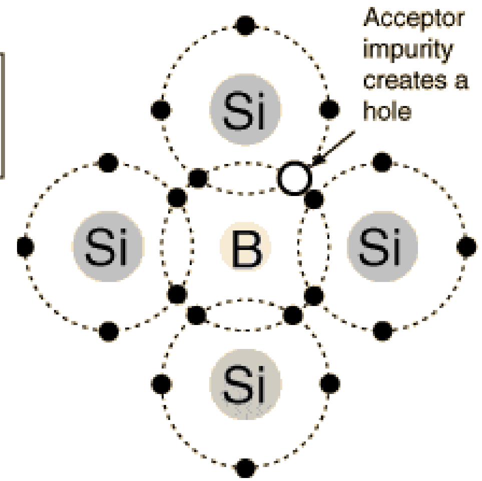
Extrinsic semiconductor:

A extrinsic semiconductor has little conductivity at room temperature. Its conductivity can be increased by the addition of a small amount of suitable metallic impurity. It is then called extrinsic semiconductor.

The process of adding impurity to a semiconductor is known as doping.



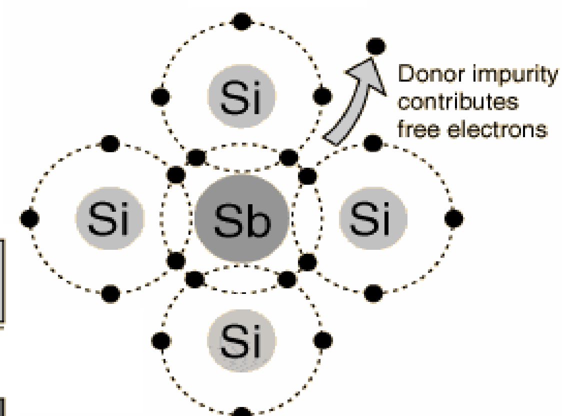




Extrinsic semiconductor:

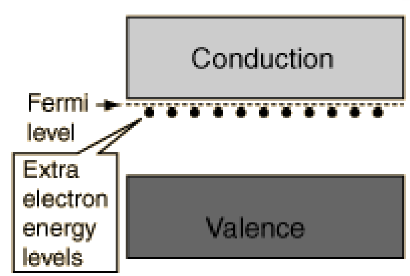
P-Type:

In P-type doping, boron or gallium is the dopant.



N-Type:

In N-type doping, phosphorus or arsenic is added to the silicon in small quantities.

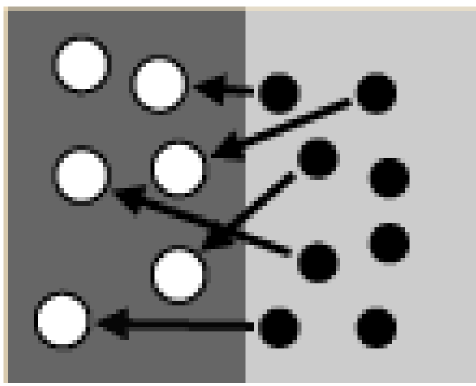


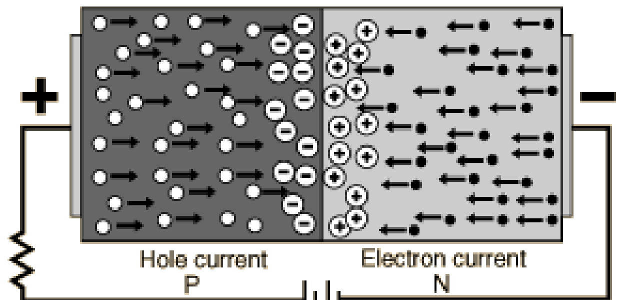
Comparison of Intrinsic and Extrinsic semiconductors

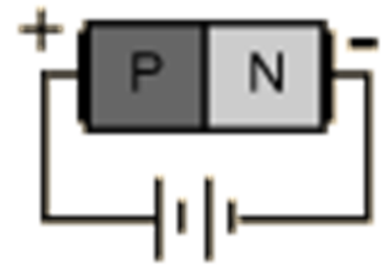
S.NO	Intrinsic Semiconductor	Extrinsic Semiconductor
1	It is in pure form.	It is formed by adding trivalent or pentavalent impurity to a pure semiconductor.
2	Holes and electrons are equal.	No. of free holes are more in p-type and no. of free electrons are more in n-type.
3	Fermi level lies in between valence and conduction Bands.	Fermi level lies near valence band in p-type and near conduction band in n-type.
4	Ratio of majority and minority carriers is unity.	Ratio of majority and minority carriers is not unity.

	p-type Semiconductor	*n*-type Semiconductor
(i)	In *p*-type semiconductor trivalent impurities are used for doping.	In *n*-type semiconductor, pentavalent impurities are used for doping.
(ii)	The impurity is called acceptor impurity.	The impurity is called donar impurity.
(iii)	In *p*-type semiconductor, the holes act as a majority charge carriers.	In *n*-type semiconductor, the electrons act as a majority charge carriers.
(iv)	In *p*-type semiconductor, conductivity is mainly due to holes, which are positively charged.	In *n*-type semiconductor, conductivity is mainly due to electrons, which are negatively charged.

PN Junction







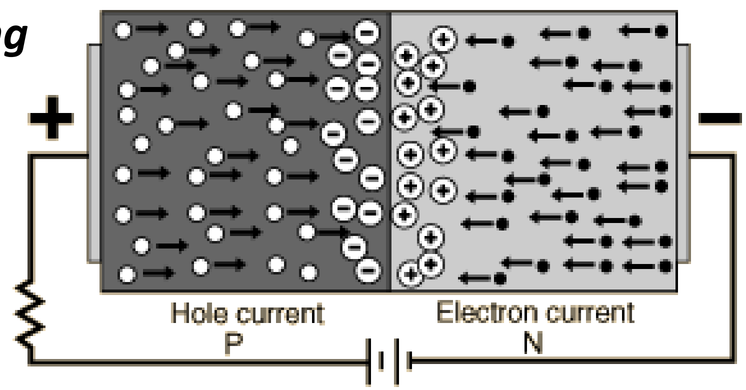
When a P-type semiconductor is suitably joined to a N-type semiconductor, the contact surface is called PN junction.

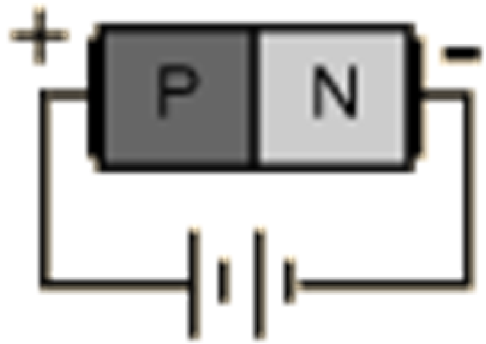
Most semiconductor devices contain one or more PN junction.

When a P-N junction is formed, some of the electrons from the N-region which have reached the conduction band are free to diffuse across the junction and combine with holes.

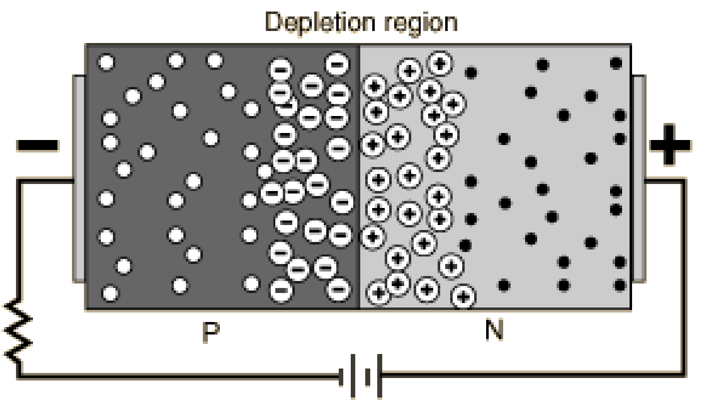
PN Junction: Reverse Biasing and Forward Biasing

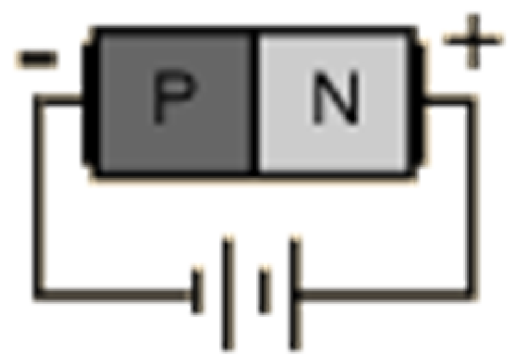
Forward biasing





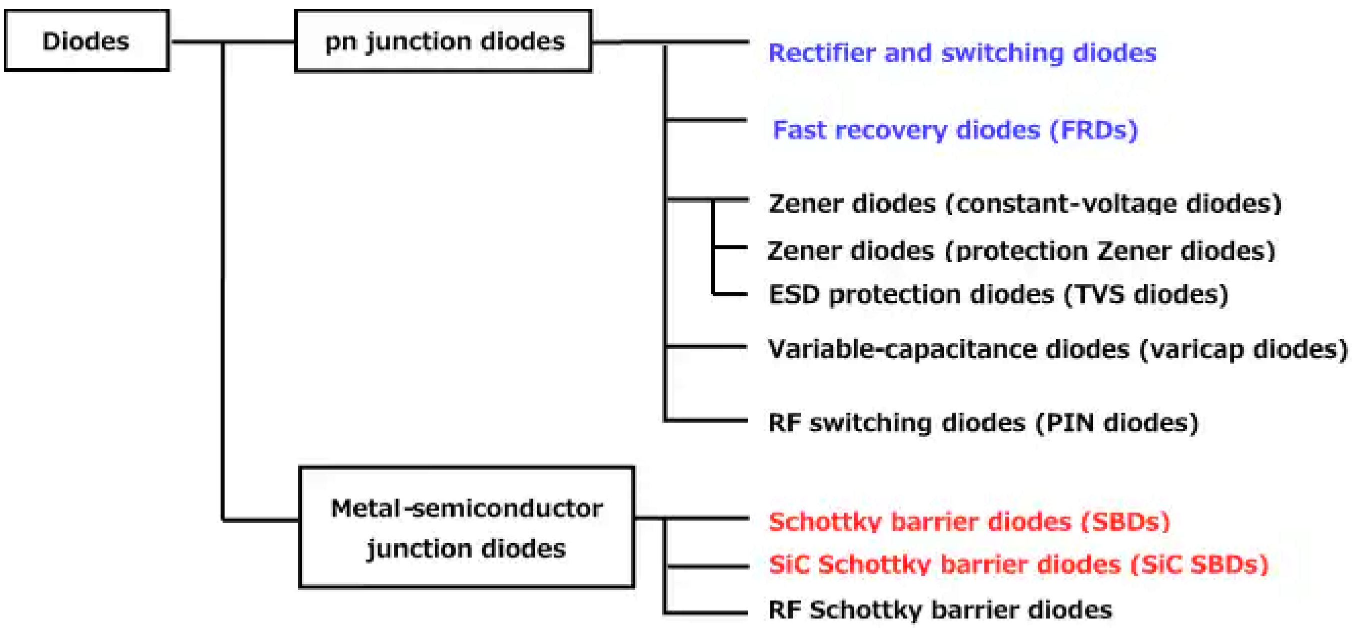
Reverse Biasing





Diode:

A diode is a two-terminal electronic component that conducts current primarily in one direction. It has low resistance in one direction, and high resistance in the other.



Diode:

A diode is a two-terminal electronic component that conducts current primarily in one direction. It has low resistance in one direction, and high resistance in the other.

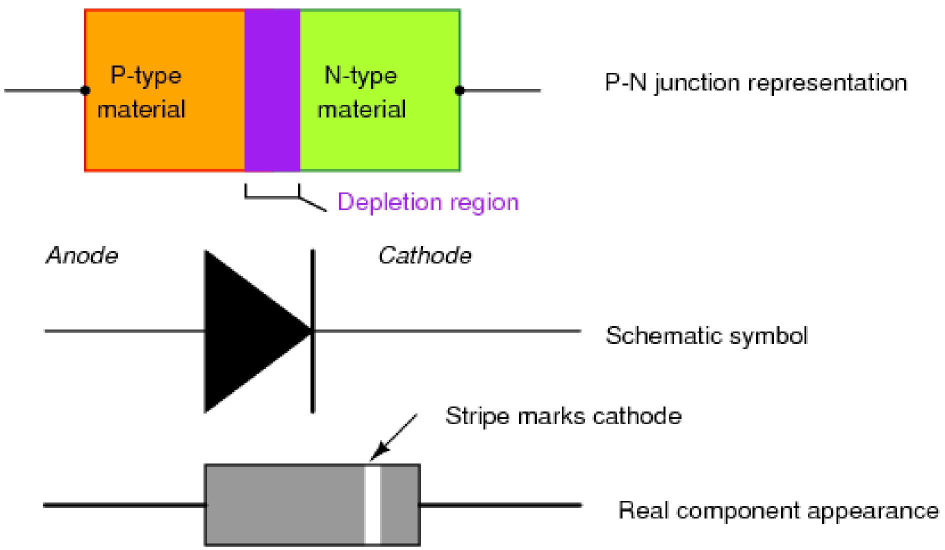
1.Rectifier and switching diode

2. Zener diode

Diode:

A diode is a two-terminal electronic component that conducts current primarily in one direction. It has low resistance in one direction, and high resistance in the other.

1.Rectifier and switching diode



Diode:

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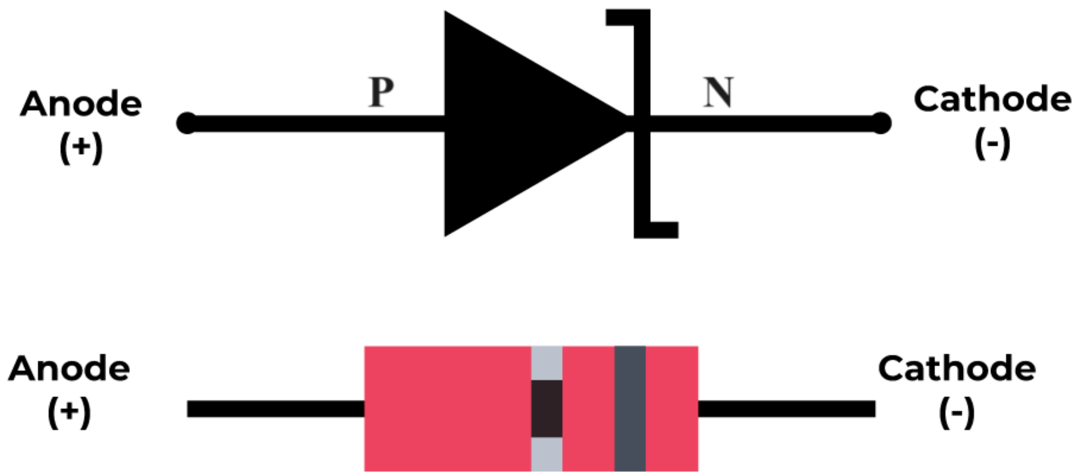
1. Zener diode



Diode applications:

- Rectifiers
- Switching diodes
- Zener diodes
- Varactor diodes (Varactor = Variable reactance)

Photodiodes

- pn junction photodiodes
- p-i-n and avalanche photodiodes

Solar Cells

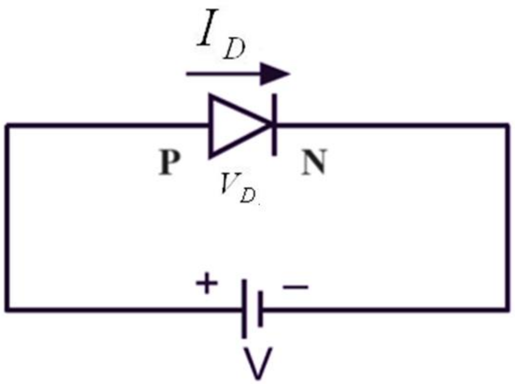
Light Emitting Diodes

Lasers

Diode Equation

The general equation linking the diode current I_D to the applied voltage V_D is:

$$I_D = I_s\left[\exp\left(\frac{eV_D}{nkT}\right) - 1\right]$$

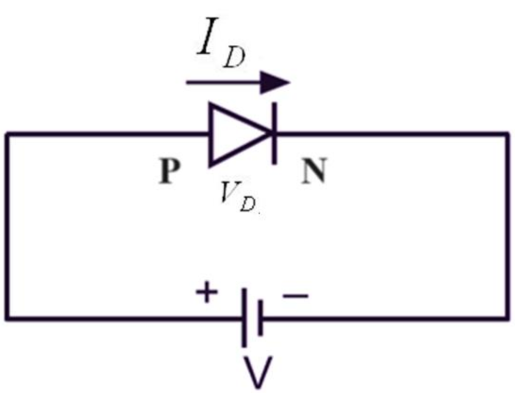


Diode Equation: $$I_D = I_s\left(e^{V_D/nV_T} - 1\right)$$

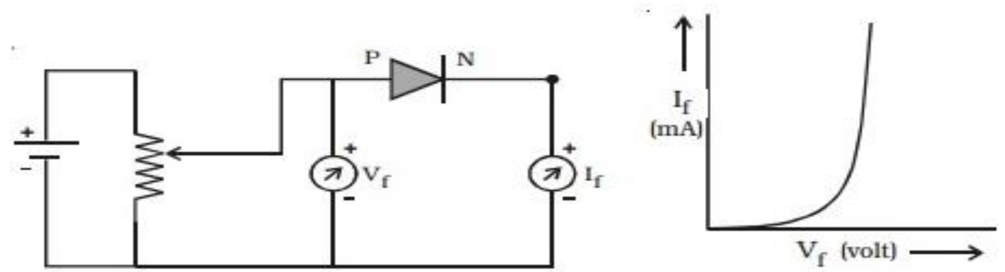
- I_D is the total diode current
- I_s reverse saturation current
- V_D applied voltage across the diode
- n an ideality factor, value between 1&2.
- V_T thermal voltage:

$$V_T = \frac{kT}{q}$$

k = 1.38 x 10^{-23} J/K
q = 1.6x10^{-19} C

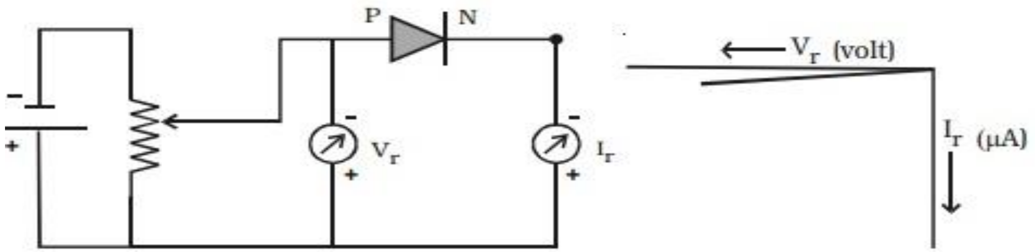


Diode Characteristics



(a) Diode circuit–Forward bias *(b) Forward characteristics*

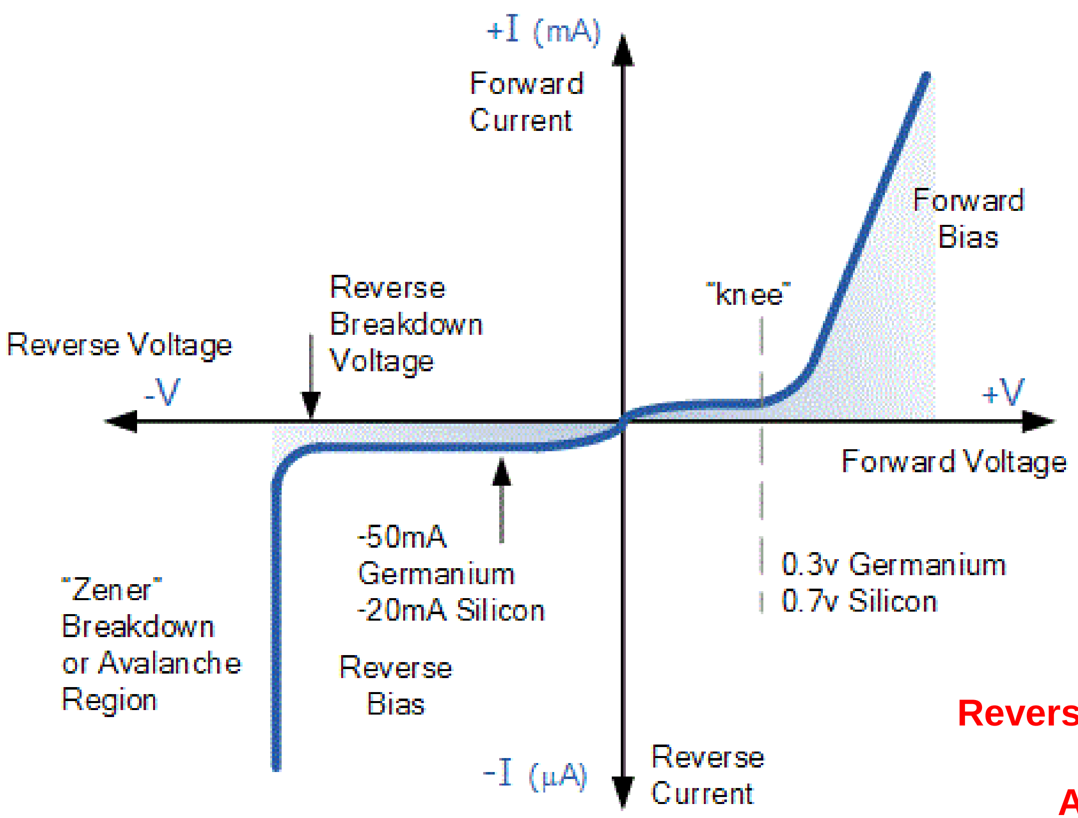
Forward bias characteristics of a diode



(a) Diode circuit–Reverse bias *(b) Reverse characteristics*

Reverse bias characteristics of a diode

Diode Characteristics



Reverse current.
Knee voltage.
Reverse Breakdown voltage.
Zener Breakdown.
Avalanche Breakdown.

Diode Characteristics

Reverse current: The current that flows in the reverse direction when the diode is reverse bias is called Reverse current. The reverse current is smaller than the forward current.

Knee voltage: The voltage at which the forward diode current increases rapidly is known as Knee voltage or cut in voltage. Knee voltage for germanium is 0.3V & for silicon is 0.7V.

Reverse Breakdown voltage: Breakdown voltage of a diode is the reverse-bias voltage at which current increases suddenly across it. If reverse bias is increased, the current through PN junction will also increase resulting in formation of a voltage called breakdown voltage.

Diode Breakdown

Avalanche Breakdown:

1. **The reverse bias increases the electric field across the depletion region.**
2. **When the high electric field exists across the depletion region, the velocity of minority charge carrier crossing the depletion region increases.**
3. **These carriers collide with the atoms of the crystal.**
4. **Because of the violent collision, the charge carrier takes out the electrons from the atom.**
5. **The collision increases the electron-hole pair. As the electron-hole induces in the high electric field, they are quickly separated and collide with the other atoms of the crystals.**
6. **The process is continuous, and the electric field becomes so much higher then the reverse current starts flowing in the PN junction. The process is known as the Avalanche breakdown.**

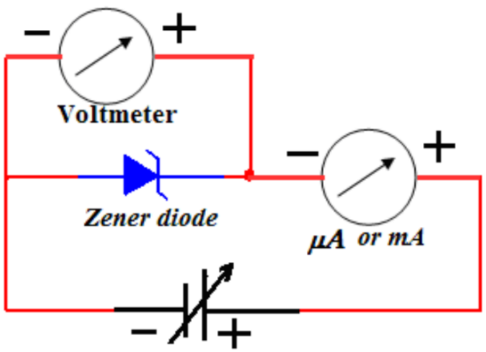
Diode Breakdown

Zener breakdown:

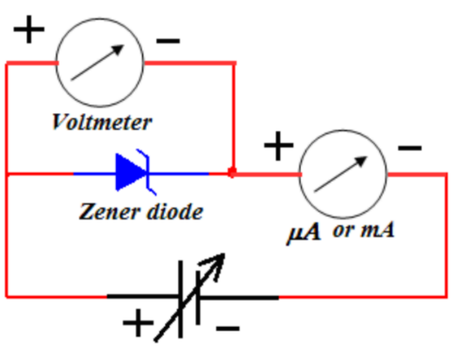
1. The phenomenon of the Zener breakdown occurs in the very thin depletion region.
2. The thin depletion region has more numbers of free electrons.
3. The reverse bias applies across the PN junction develops the electric field intensity across the depletion region.
4. The strength of the electric field intensity becomes very high.
5. The electric field intensity increases the kinetic energy of the free charge carriers.
6. Thereby the carriers start jumping from one region to another.
7. These energetic charge carriers collide with the atoms of the p-type and n-type material and produce the electron-hole pairs.
8. The reverse current starts flowing in the junction because of which depletion region entirely vanishes. This process is known as the Zener breakdown.

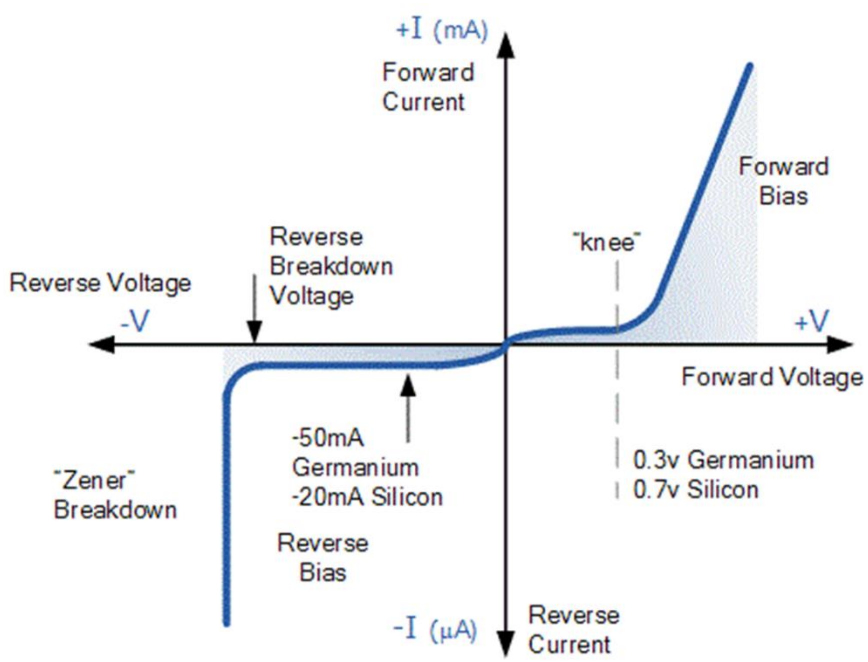
Zener Diode Characteristics

Zener Diode in RB



Zener Diode in FB





Applications of Zener Diode

- **As a voltage regulator**
- **Protects from overvoltage**
- **Used in clipping circuits**
- **Used to shift voltage**

Rectifiers

What is a rectifier?

A rectifier is an electronic circuits that converts an alternating current into a pulsating direct current by using one or more P-N junction diodes.

What are the types of rectifiers?

The following are the types of rectifiers:

- **Uncontrolled Rectifier**
- **Controlled Rectifier**

An uncontrolled rectifier is a type of AC to DC converter that provides fixed DC output voltage from a given AC input supply. These rectifiers are used in applications where fixed DC voltage is essential.

A controlled rectifier can control the power fed to the load. It is used to convert AC supply into unidirectional DC supply in an inverter. Controlled rectification is the process of converting AC to direct current (DC) based on the required voltage and current demand.

Rectifiers

Uncontrolled Rectifiers

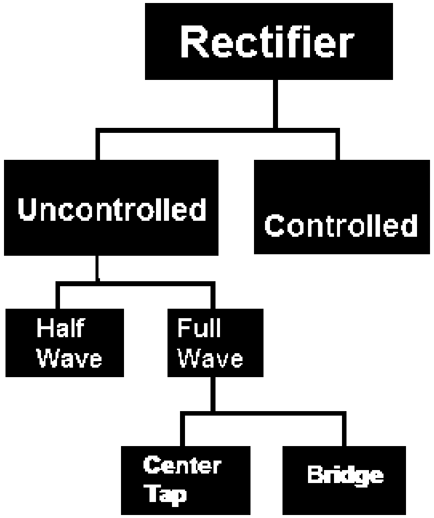
The type of rectifier whose voltage cannot be controlled is known as an uncontrolled rectifier. Uncontrolled rectifiers are further divided as follows:

Half Wave Rectifier
Full Wave Rectifier

The two most common types of full wave rectifiers are the

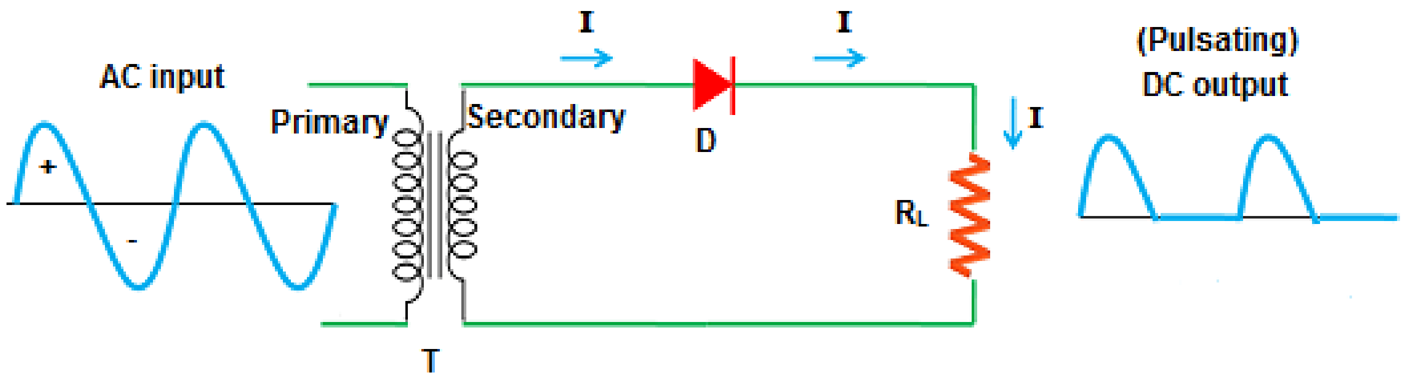
(i) Bridge rectifier
(ii) Center-tapped rectifier.

The bridge rectifier uses four diodes to form a bridge circuit, while the center-tap rectifier uses a transformer with a center-tapped secondary winding and two diodes.



Rectifiers

Half-Wave Rectifiers



I = Current

D = Diode

R_L = Load resistor

T = Transformer

+ = Positive half cycle

- = Negative half cycle

Half wave rectifier

Rectifiers

Half-Wave Rectifiers

$$Average\ Output\ Current$$

$$= \left(\frac{1}{2\pi}\right)\left[\int_0^{\pi} ImSin\omega td(wt) + \int_{\pi}^{2\pi} 0d(\omega t)\right]$$

$$= \left(\frac{1}{2\pi}\right)\int_0^{\pi} ImSin\omega td(wt)$$

$$= \left(\frac{Im}{2\pi}\right)\int_0^{\pi} Sin\omega td(wt)$$

$$= \left(\frac{-Im}{2\pi}\right)[cos\pi - cos0]$$

$$= \frac{Im}{\pi}$$

$$RMS\ Value = \sqrt{\left(\frac{1}{2\pi}\right)\int_0^{\pi}(Imsin\omega t)^2 d(\omega t) + \left(\frac{1}{2\pi}\right)\int_{\pi}^{2\pi} 0\, d\omega t}$$

$$= \sqrt{\left(\frac{Im^2}{2\pi}\right)\int_0^{\pi}(sin\omega t)^2 d(\omega t)}$$

$$= Im\sqrt{\left(\frac{1}{4\pi}\right)\int_0^{\pi} 2(sin\omega t)^2\, d(\omega t)}$$

$$= \left(\frac{Im}{2}\right)\sqrt{\left(\frac{1}{\pi}\right)\int_0^{\pi}(1 - cos\omega t)\, d(\omega t)}$$

$$= \left(\frac{Im}{2}\right)\sqrt{\left(\frac{1}{\pi}\right)\left[\int_0^{\pi} 1d(\omega t) - \int_0^{\pi} cos\omega t\, d(\omega t)\right]}$$

$$= \left(\frac{Im}{2}\right)\sqrt{\left(\frac{1}{\pi}\right)[\pi - 0)]}$$

$$= \frac{Im}{2}$$

Rectifiers

Half-Wave Rectifiers

Ripple factor:
The ripple factor is **the ratio of the RMS value of AC component present in rectifier Output to the DC voltage of the Rectifier Output**. The ripple factor is denoted as γ. It is a dimensionless quantity and always has a value less than unity.

$$I_{rms}^2 = I_{dc}^2 + I_{ac}^2$$

$$I_{rms} = \sqrt{I_{dc}^2 + I_{ac}^2}$$

or $$I_{ac} = \sqrt{I_{rms}^2 - I_{dc}^2}$$

Dividing throughout by I_{dc}, we get,

$$\frac{I_{ac}}{I_{dc}} = \frac{1}{I_{dc}}\sqrt{I_{rms}^2 - I_{dc}^2}$$

But I_{ac}/I_{dc} is the ripple factor.

$$\therefore \text{Ripple factor} = \frac{1}{I_{dc}}\sqrt{I_{rms}^2 - I_{dc}^2} = \sqrt{\left(\frac{I_{rms}}{I_{dc}}\right)^2 - 1}$$

Rectifiers

Full Wave Rectifier:

A full wave rectifier is defined as **a rectifier that converts the complete cycle of alternating current into pulsating DC**. Unlike half-wave rectifiers that utilize only the halfwave of the input AC cycle, full wave rectifiers utilize the full cycle.

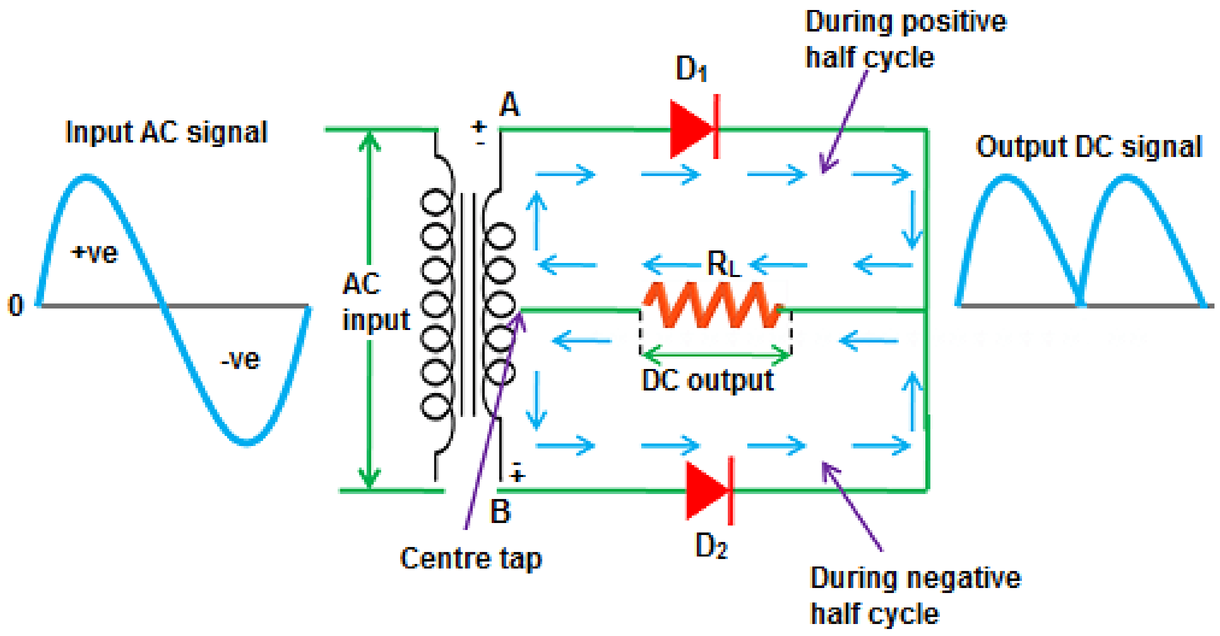
Center-tapped full wave rectifier: A center-tapped full wave rectifier is a type of rectifier that uses a center-tapped transformer and two diodes to convert the complete AC signal into DC signal.

Bridge rectifier: A diode bridge is a bridge rectifier circuit of four diodes that is used in the process of converting alternating current from the input terminals to direct current on the output terminals.

Rectifiers

Full Wave Rectifier:

Center-tapped full wave rectifier: A center-tapped full wave rectifier is a type of rectifier that uses a center-tapped transformer and two diodes to convert the complete AC signal into DC signal.



Rectifiers

Full Wave Rectifier:

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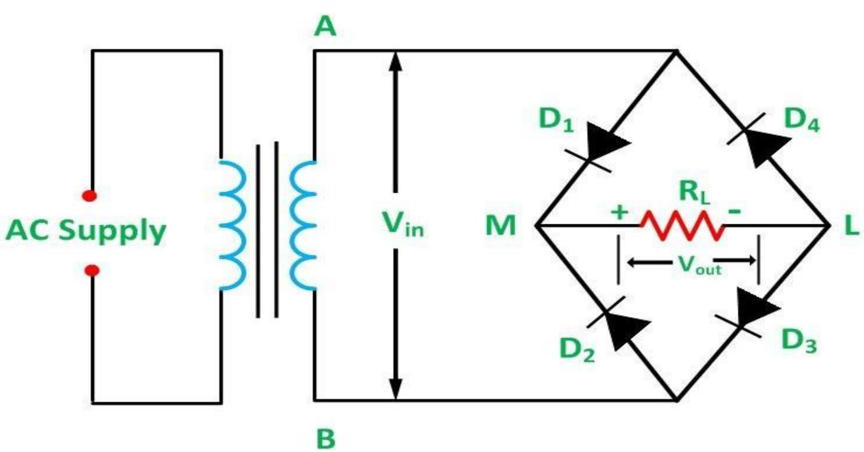
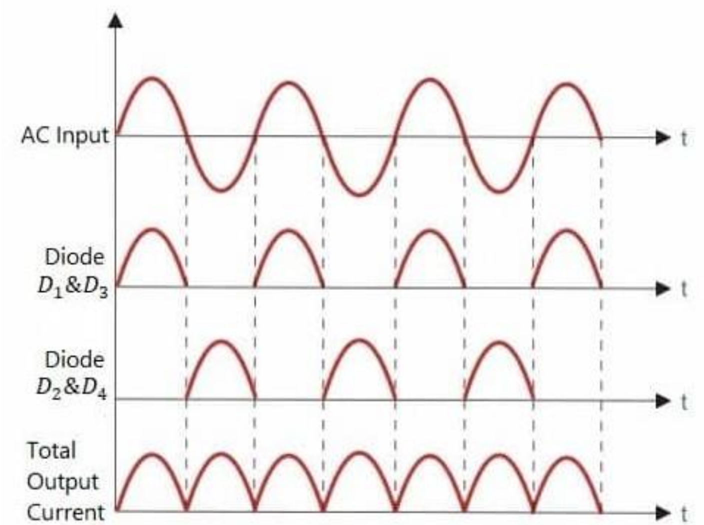




Input Output waveform of Full Wave Bridge Rectifier

Rectifiers

Full Wave Rectifier:

R.M.S. Value

Average value

Ripple factor

$$Average\ Value = \frac{1}{\pi}\int_0^{\pi} Imsin\omega td(wt)$$

$$= \frac{Im}{\pi}\int_0^{\pi} sin\omega td(wt)$$

$$= \left(\frac{Im}{\pi}\right)[cos0 - cos\pi]$$

$$= (2\frac{Im}{\pi})$$

$$= 0.637Im$$

$$RMS\ Value = \sqrt{\frac{1}{\pi}\int_0^{\pi}(Imsin\omega t)^2 d(\omega t)}$$

$$= Im\sqrt{\frac{1}{2\pi}\int_0^{\pi} 2(sin\omega t)^2 d(\omega t)}$$

$$= Im\sqrt{\frac{1}{2\pi}\int_0^{\pi}(1 - cos\omega t)d(\omega t)}$$

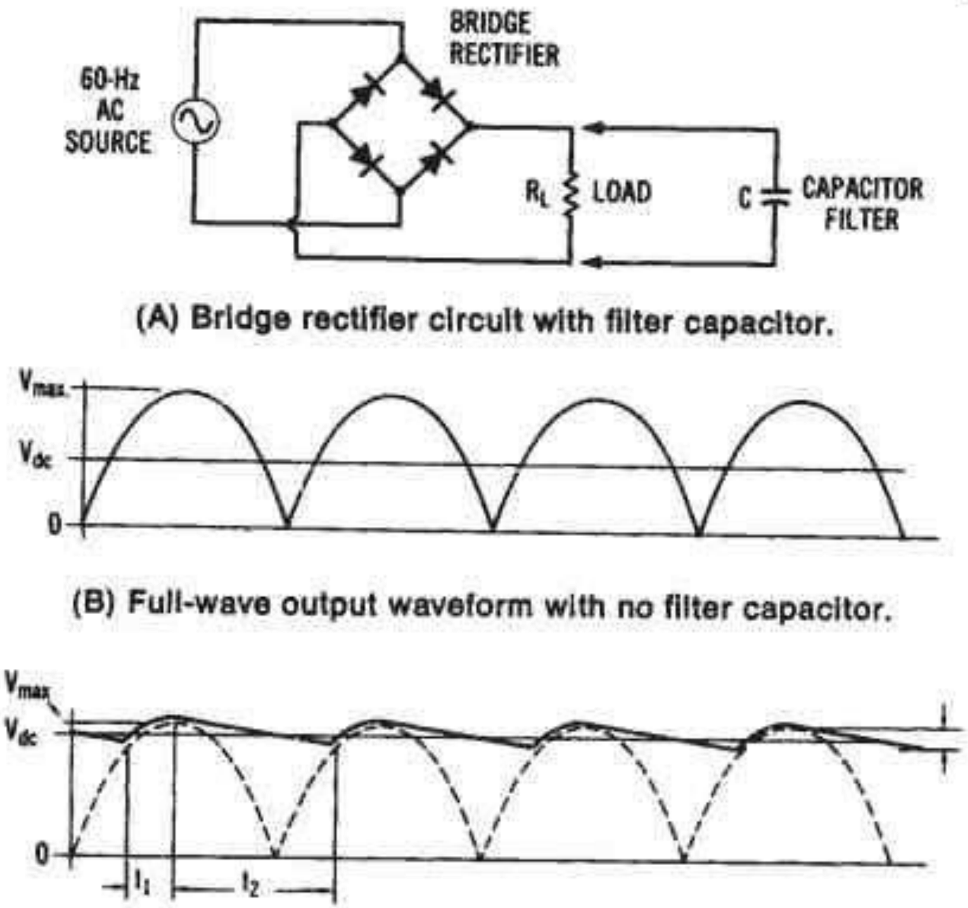
$$= Im\sqrt{\left(\frac{1}{2\pi}\right)[\pi - 0]}$$

$$= \frac{Im}{\sqrt{2}}$$

$$I_{rms} = \frac{I_m}{\sqrt{2}} \qquad I_{dc} = \frac{2\,I_m}{\pi}$$

$$\text{Ripple factor} = \sqrt{\left(\frac{I_m/\sqrt{2}}{2\,I_m/\pi}\right)^2 - 1} = 0.48$$

Rectifier with filter and without filter



(A) Bridge rectifier circuit with filter capacitor.

(B) Full-wave output waveform with no filter capacitor.

What is a Filter?

Difference between half wave and full wave rectifier

Half Wave Rectifier	Full Wave Rectifier
1) Only one diode is used.	1) Two diodes are used.
2) The output is discontinuous and pulsative d.c.	2) The output is continuous and pulsating d.c.
3) Efficiency is 40.6%.	3) Efficiency is 81.2%.
4) Only one half of the a.c. input wave comes out as d.c. output	4) Both half cycles of a.c. input wave comes out as d.c. output.
5) Efficiency is low.	5) Efficiency is high.

Difference between Full Wave Center tap and Bridge Rectifier

Properties	Full Wave Center Tap Rectifier	Full Wave Bridge Rectifier
Number of Diodes	2	4
D.C Current	2 Im / π	2 Im / π
Transformer Necessary	Yes	No
Max Value of Current	Vm / (rf + RL)	Vm / (2rf + RL)
Ripple Factor	0.482	0.482
O/P Frequency	2 fin	2 fin
Max Efficiency	81.2%	81.2%
Peak Inverse Voltage	2 Vm	2 Vm

Voltage Regulation

Voltage regulation describes the ability of a system to provide near constant voltage over a wide range of load conditions.

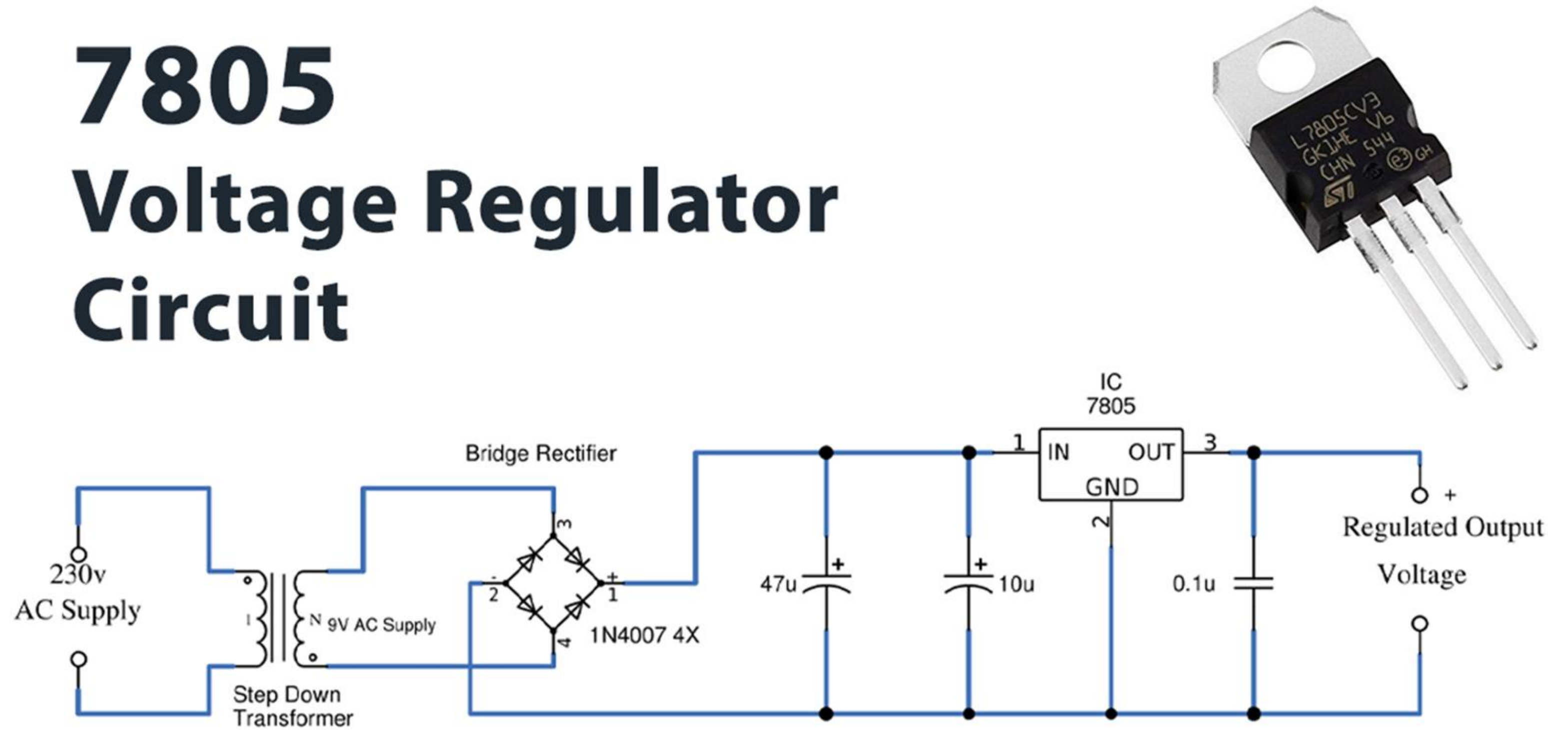
A voltage regulator is a system designed to automatically maintain a constant voltage.

A voltage regulator is **an electronic circuit that provides a stable DC voltage independent of the load current, temperature and AC line voltage variations**.

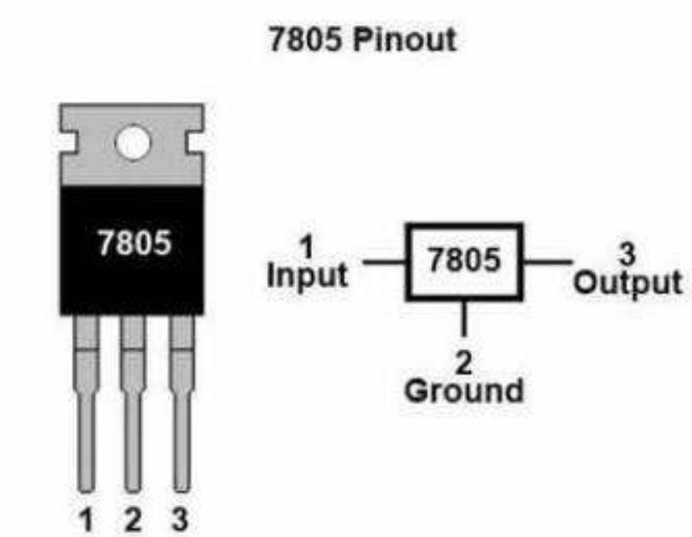
Voltage regulators can be classified into two types:

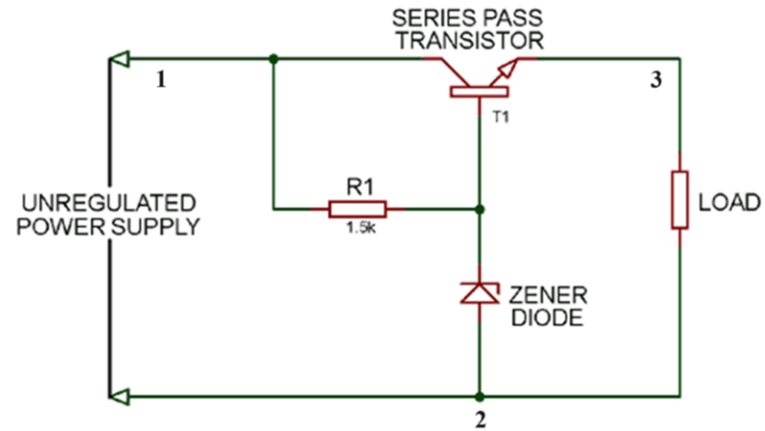
- Linear Voltage Regulators
- Switching Voltage Regulators

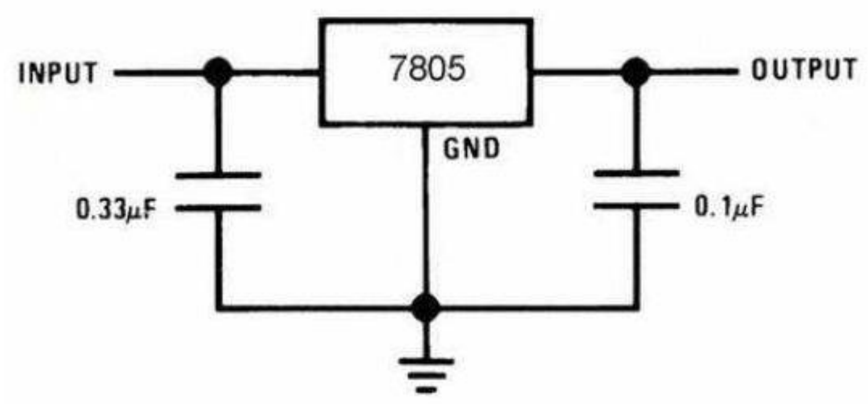
Voltage Regulation



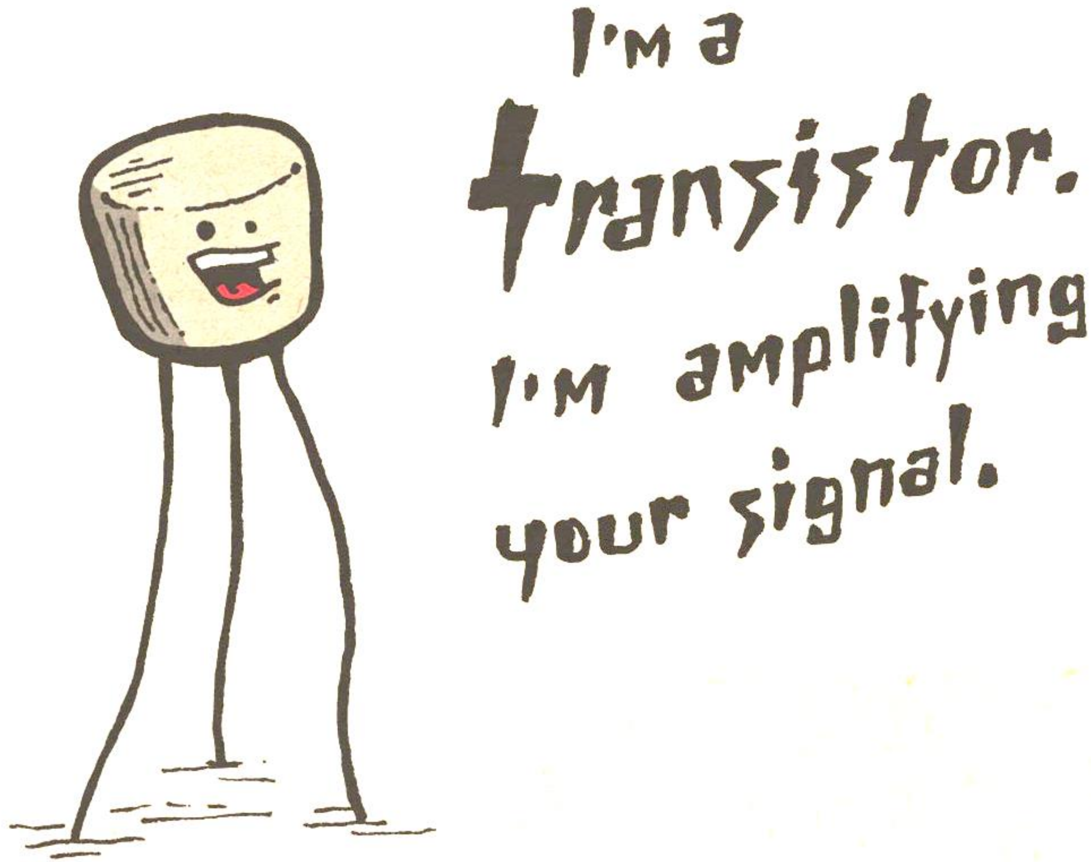
Voltage Regulation





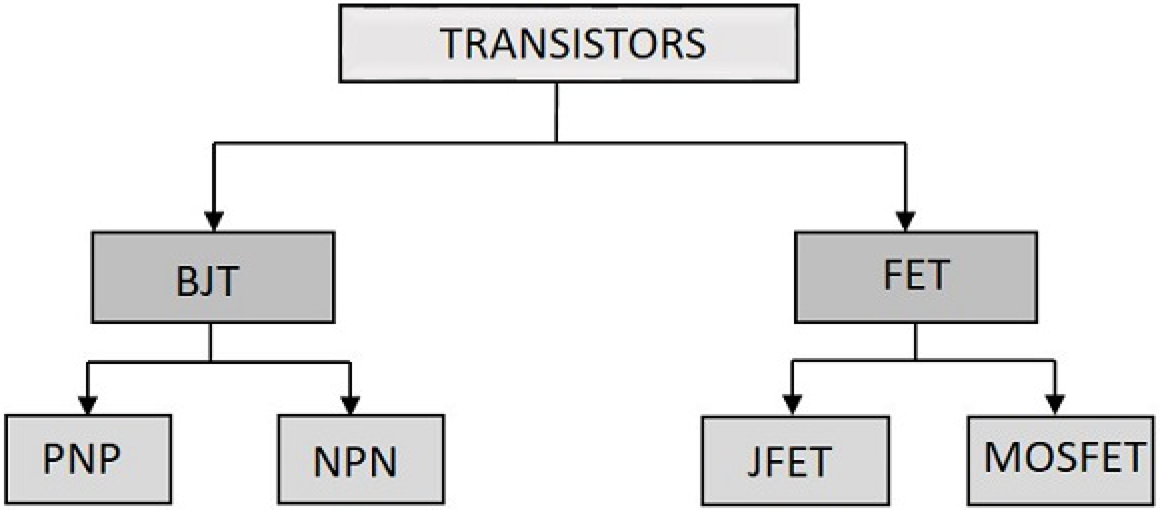


Transistor



Transistor

A transistor is a three terminal semiconductor device used to amplify or switch electrical signals and power. It is one of the basic building blocks of modern electronics.

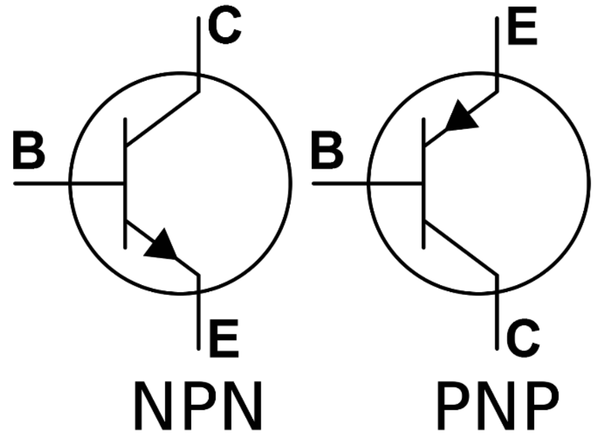


Transistor

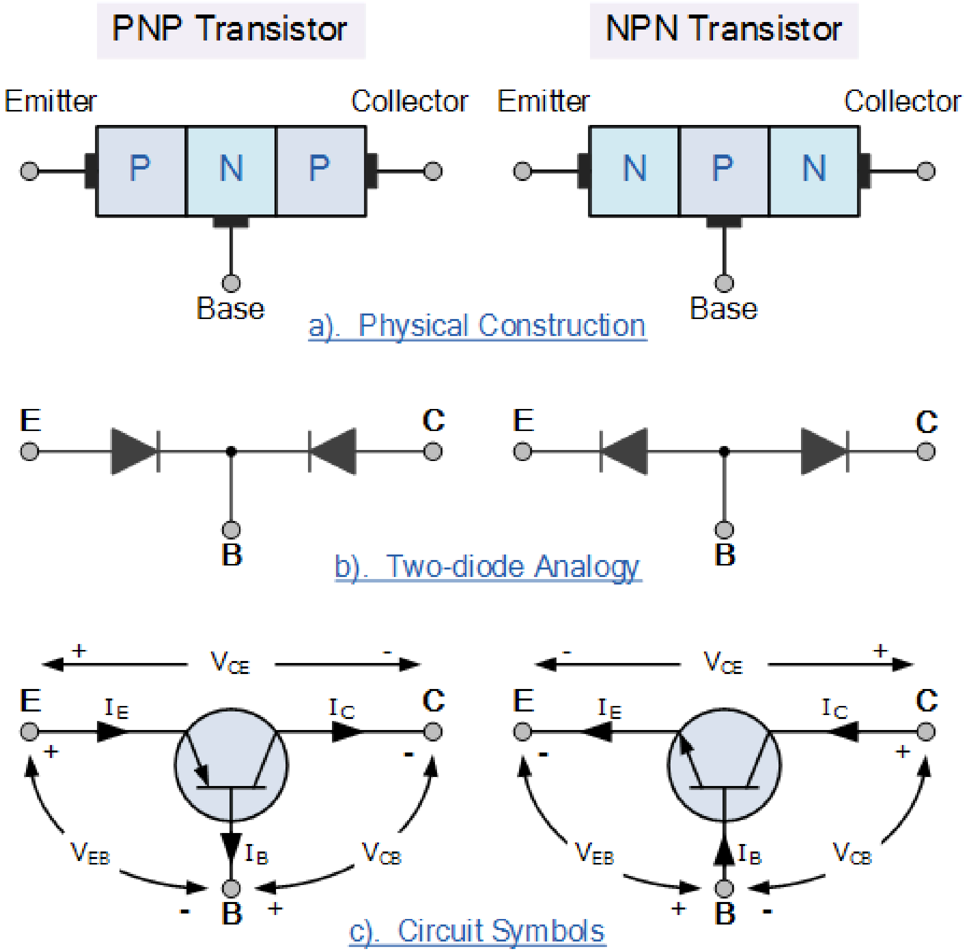
A transistor is a three terminal semiconductor device used to amplify or switch electrical signals and power. It is one of the basic building blocks of modern electronics.

A bipolar junction transistor is a type of transistor that uses both electrons and electron holes as charge carriers. In contrast, a unipolar transistor, such as a field-effect transistor, uses only one kind of charge carrier.

A bipolar transistor (bipolar junction transistor: BJT) consists of three semiconductor regions forming two junctions. There are two types of structure: NPN and PNP.



Transistor

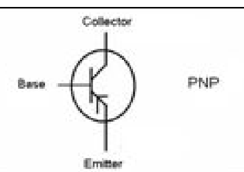
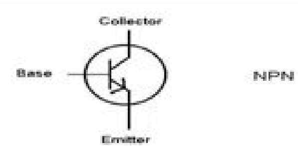


a). Physical Construction

b). Two-diode Analogy

c). Circuit Symbols

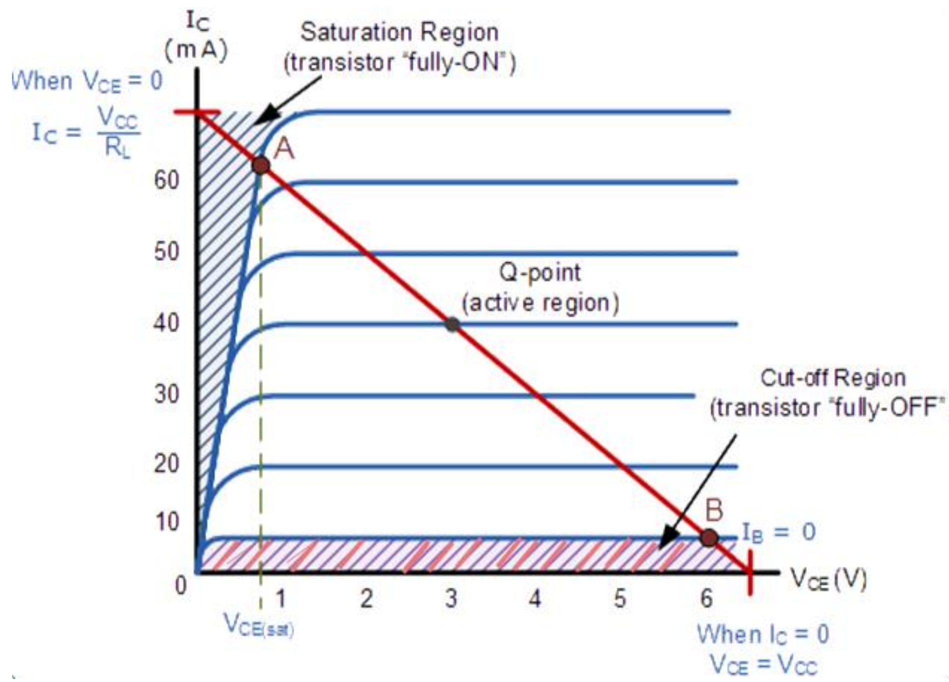
Differences between NPN & PNP

Type of BJT	PNP-Type	NPN-Type
1	If the base is at a lower voltage than the emitter, current flows from emitter to collector	If the base is at a higher voltage than the emitter, current flows from collector to emitter.
2	Small amount of current also flows from emitter to base.	Small amount of current also flows from base to emitter.
3	Emitter is heavily p-doped compared to collector. So, emitter and collector are not interchangeable.	Emitter is heavily N-doped compared to collector. So, emitter and collector are not interchangeable.
4	The base width is small compared to the minority carrier diffusion length. If the base is much larger, then this will behave like back-to-back diodes.	The base width is small compared to the minority carrier diffusion length. If the base is much larger, then this will behave like back-to-back diodes.
5	Voltage at base controls amount of current flow through transistor (emitter to collector).	Voltage at base controls amount of current flow through transistor (collector to emitter).
6		
7	Follow the arrow to see the direction of current flow	Follow the arrow to see the direction of current flow

Transistor

Transistor Regions of Operation

Operation Region	Bias	Application
CUT OFF REGION	I_3 and I_c are 0 (base-emitter junction is reverse biased)	Open Switch (OFF)
SATURATION REGION	Base emitter junction is forward biased; I_3 flows	Closed Switch (ON)
ACTIVE REGION	B-E junction is forward biased but C-E junction is reverse biased;	Amplifier

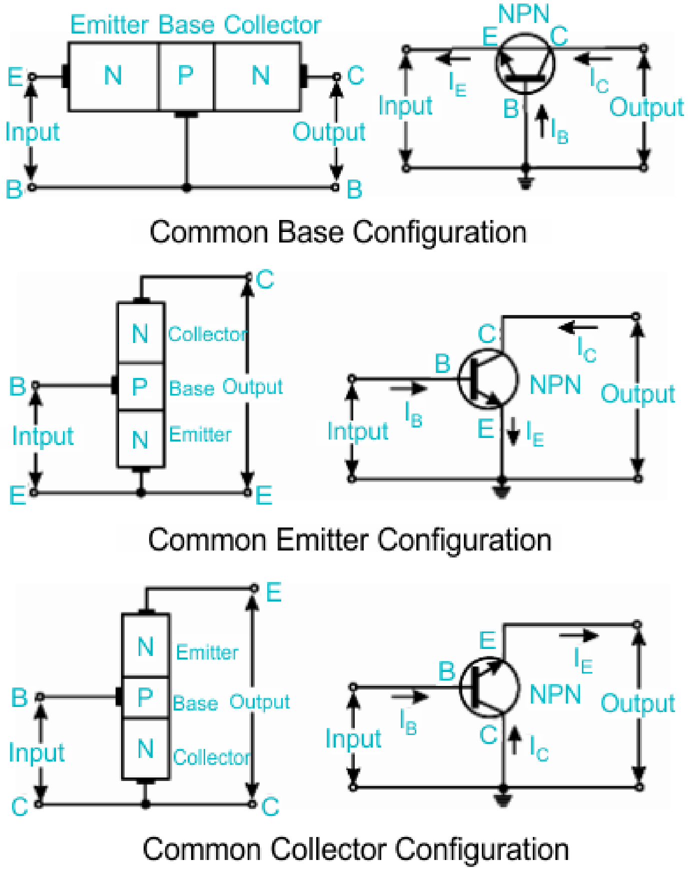


Transistor

Types of Transistor Configurations

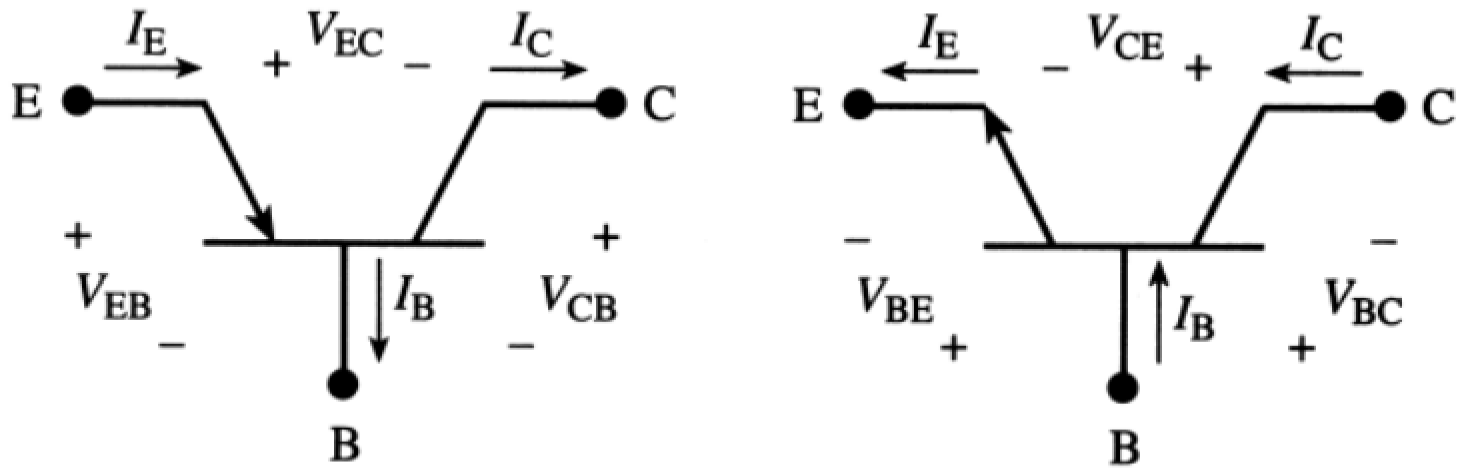
Transistor can be connected in a circuit in the following ways.

1. **Common base (CB) configuration (or) Grounded base configuration,**
2. **Common emitter (CE) configuration (or) Grounded emitter configuration.**
3. **Common collector (CC) configuration (or) Grounded collector configuration.**



Transistor

Current Equations



(a) *pnp*

(b) *npn*

$$I_E = I_C + I_B$$

Transistor

BJT Characteristics (CE)

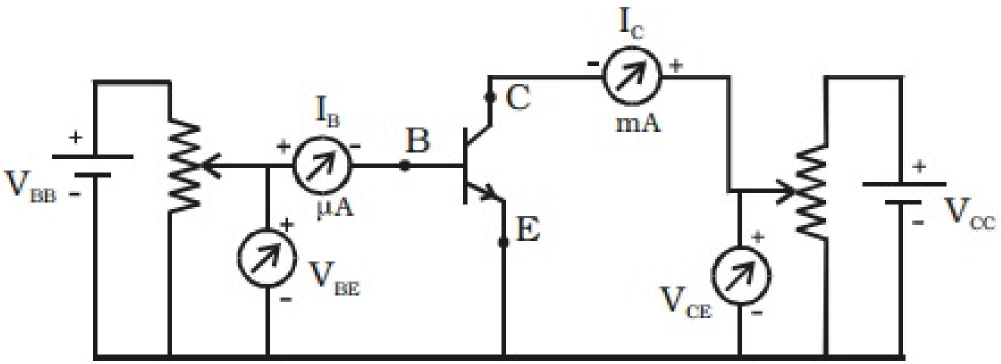
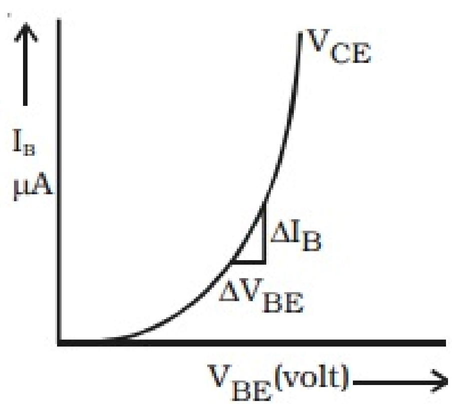
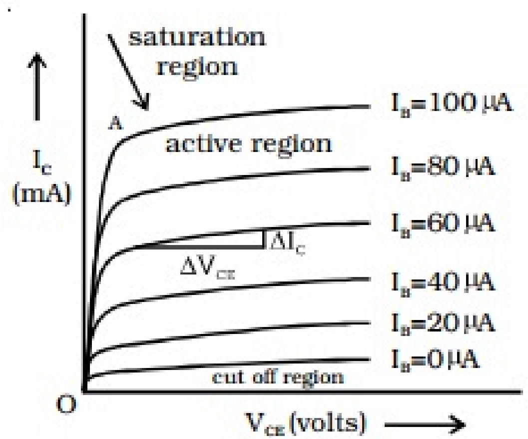


Fig Transistor circuit in CE mode.



Input characteristics



Output characteristics

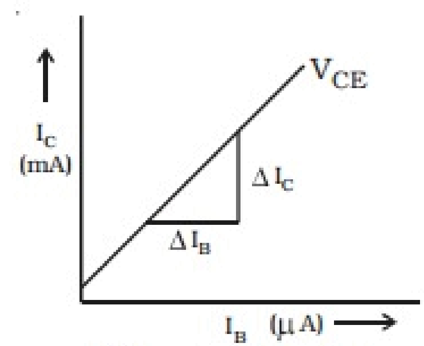
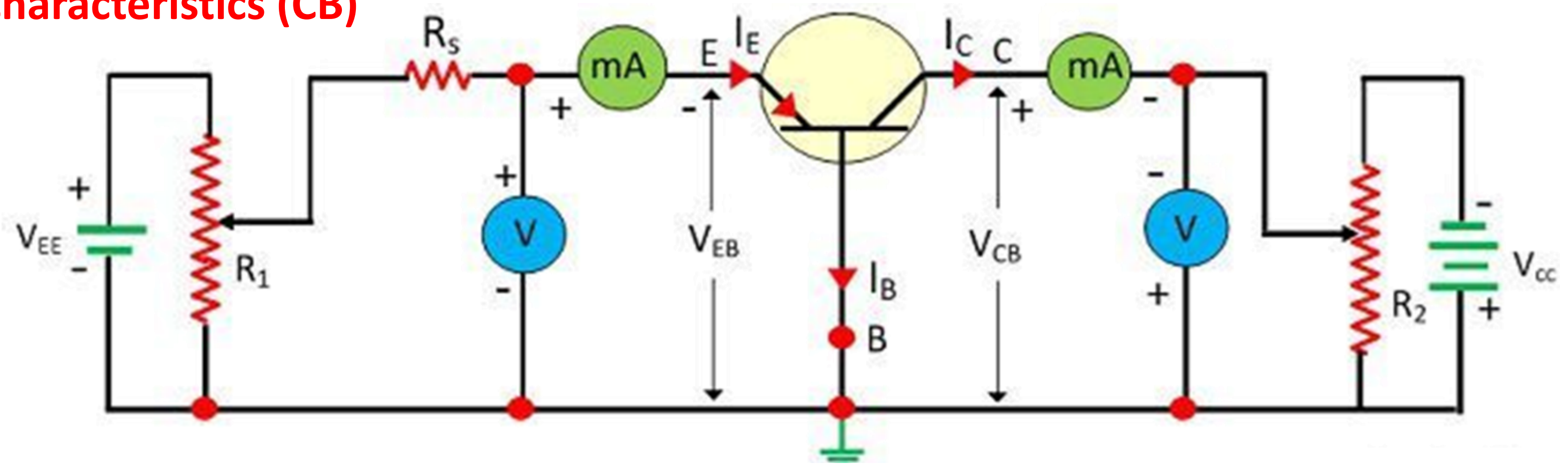
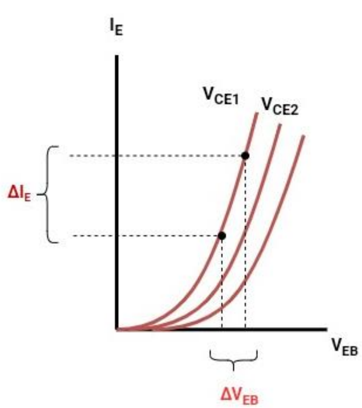


Fig Transfer characteristic curve

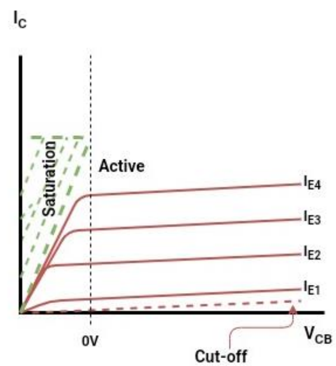
Transistor

BJT Characteristics (CB)





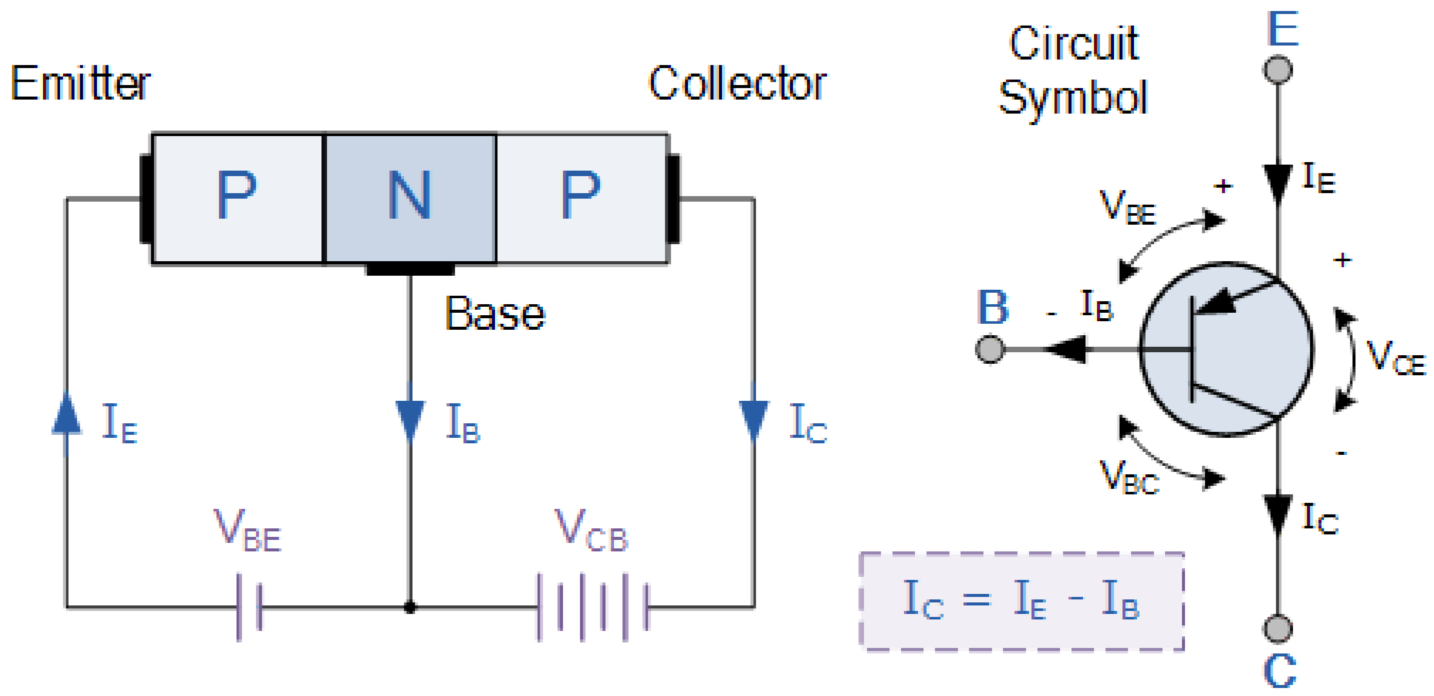
(c) Input characteristics



(d) Output characteristics

Transistor

Current Equations: $I_E = I_C + I_B$



Transistor

Current Equations: $I_E = I_C + I_B$

$$I_E = I_B + I_C$$

CB Mode

CE Mode

CC Mode

$$\text{Alpha}(\alpha) = \frac{I_C}{I_E} \quad \text{and} \quad \text{Beta}(\beta) = \frac{I_C}{I_B}$$

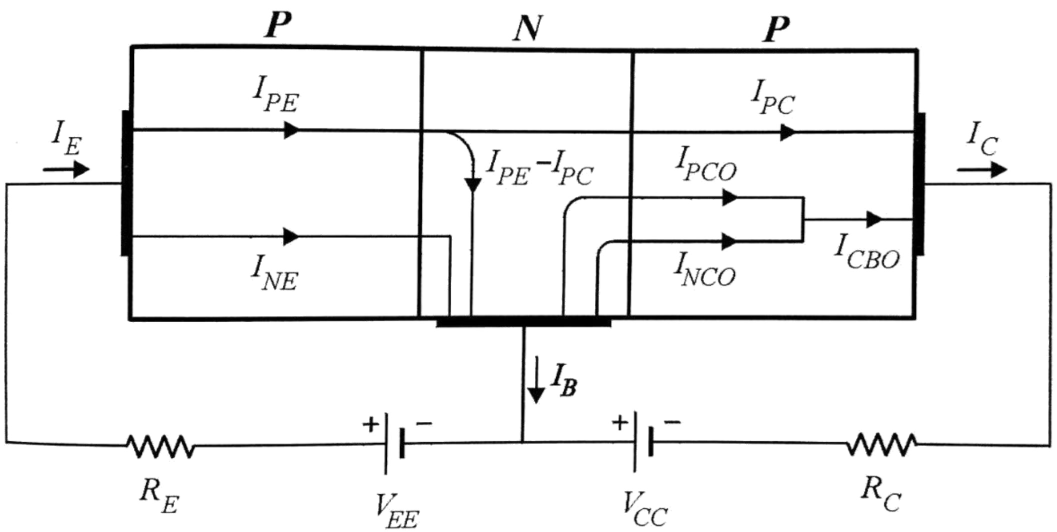
$$\gamma = \frac{1}{1-\alpha}$$

$$\therefore I_C = \alpha \times I_E = \beta \times I_B$$

$$\text{thus: } \alpha = \frac{\beta}{\beta + 1} \qquad \beta = \frac{\alpha}{1 - \alpha}$$

Transistor

Current Equations: $I_E = I_C + I_B$



$$I_C = \alpha I_\beta + I_{CBO}$$

$$I_E = I_C + I_B = (\alpha I_\beta + I_{CBO}) + I_B$$

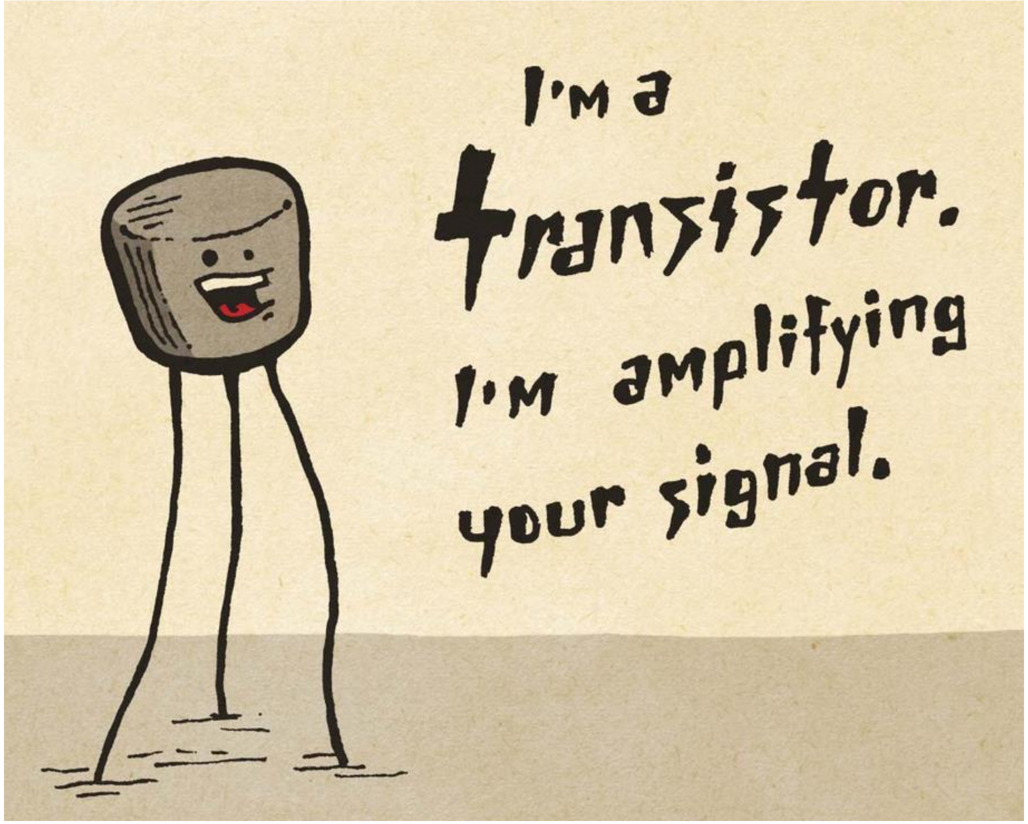
$$I_E(1-\alpha) = I_\beta + I_{CBO}$$

$$I_E = I_B\left(\frac{1}{1-\alpha}\right) + I_{CBO}\left(\frac{1}{1-\alpha}\right)$$

$$= (\beta + 1)I_B + (\beta + 1)I_{CBO}$$

Transistor

Transistor as an Amplifier

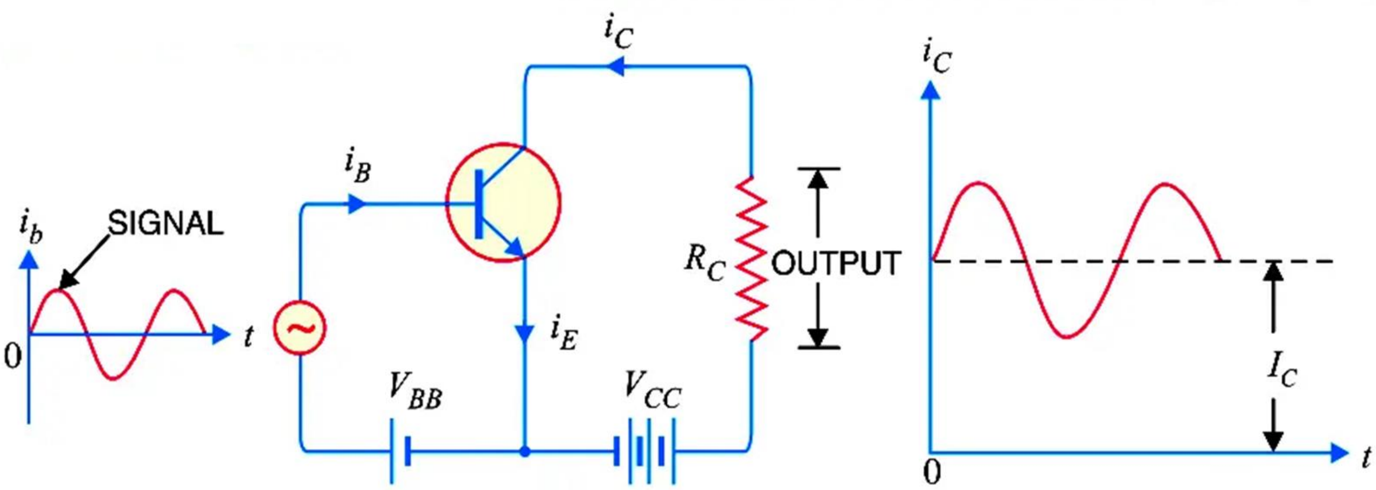


Transistor

Transistor as an Amplifier

An amplifier is *an electronic device that increases the voltage, current, or power of a signal*.

$$v_{CE} = V_{CC} - i_C R_C$$

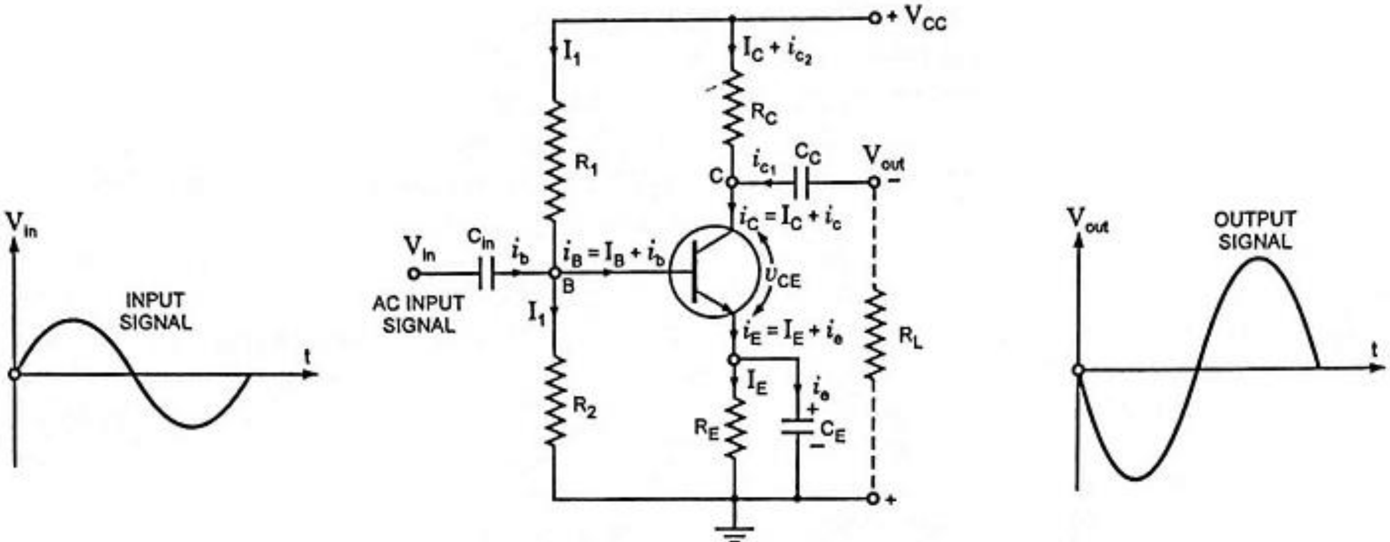


Transistor

Transistor as a Small Signal Amplifier

An amplifier is *an electronic device that increases the voltage, current, or power of a signal*.

$$v_{CE} = V_{CC} - i_C R_C$$



Practical Circuit of a Transistor Amplifier in CE Configuration With Self-Biasing

THANK YOU